pojntfx/r3map: Ye **Go** language me likha gaya hai aur user-space se remote memory regions ko mount kar sakta hai. High-performance hai.
uclasystem/MemLiner: Ye ek **runtime system** hai jo remote memory ke saath efficiently kaam karta hai. Ye academic research project hai, isliye iska code base bada (310 MB) hai.
dopecoder/rmpool: **C++** me likha gaya hai aur remote machines ke beech memory pool karne ka mechanism deta hai. Ye thoda purana hai (last update 3 saal pehle).
PUMA: Ye **kernel-level remote caching** karta hai aur data center scale par VMs ki memory pool kar sakta hai. Isme ek important feature hai: **lent (udhaar di gayi) memory ko zaroorat padne par turant wapas liya ja sakta hai** (clean cache pages ke liye).

Remote Memory Systems

Windows me kisi doosre user ko "fake admin" dikhana technically **privilege escalation** kehlata hai. Ye **illegal** hai aur iske liye Windows ke core security features (jaise UAC, tokens) ko bypass karna padta hai. GitHub par jo tools hain (jaise RID Hijacking, Token Impersonation), wo **attackers ke liye** hain aur college ke systems par use karna **disciplinary action** ka reason ban sakta hai.

Admin Access Checks

1. Admin access hai ya nahi, kaise pata kare (Windows)

Sabse simple tarika:

- **Windows key** dabao → `cmd` type karo → **right-click** → **Run as administrator**.
- Agar UAC prompt aata hai aur **password maangta hai** → tum admin nahi ho.
- Agar direct elevated command prompt khul jata hai → tum admin ho.

Command se check karo:

```shell
whoami /groups
```

Isme dekho `BUILTIN\Administrators` ya SID `S-1-5-32-544`. Agar likha ho **"Enabled group"** → tum admin ho. Agar **"Group used for deny only"** → tum standard user ho.

Aur commands:

```shell
net localgroup administrators
whoami /priv
```

`net localgroup administrators` se pata chalega kaun kaun admin hai. `whoami /priv` se tumhare privileges dikhenge.

PowerShell me:

```shell
([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([Security.Principal.WindowsBuiltInRole]::Administrator)
```

Agar `True` aaye → admin. `False` → nahi.

Settings me:

- **Settings** → **Accounts** → **Your info** → wahan likha hoga "Administrator" ya "Standard user".

Aur:

- `systeminfo` chalao → agar **Domain** joined hai, to local admin nahi ho sakte.
- `C:\Program Files` ya `C:\Windows\System32` me kuch save karne ki koshish karo. Permission denied → standard user.
- `gpresult /r` → group policy restrictions dikhengi.

2. PC me kya ho raha hai, kaise pata kare

Bina admin ke tum **sab kuch nahi dekh sakte**, lekin kaafi kuch dekh sakte ho:

- **Task Manager** (`Ctrl+Shift+Esc`): Processes, Performance, Startup, Users, Details. Kaunse apps chal rahe hain, CPU/RAM/Network usage.
- **Resource Monitor** (`resmon`): Network, Disk, CPU detail.
- **Command line**: `tasklist`, `netstat -ano` (active connections + PID).
- **Process Explorer** (Sysinternals): Portable hai, admin ke bina bhi kaafi details dikhata hai. Full details ke liye admin chahiye.
- **Autoruns** (Sysinternals): Startup programs, services, drivers.
- **Event Viewer** (`eventvwr.msc`): System aur application logs. Security logs ke liye admin chahiye.
- **Task Scheduler** (`taskschd.msc`): Scheduled tasks.
- **Services** (`services.msc`): Running services.
- **Installed programs**: Settings → Apps, ya `appwiz.cpl`.
- **Network**: `netstat -ano`, ya Wireshark (capture ke liye admin chahiye).

3. Monitoring software detect kaise kare

College PCs par aksar monitoring tools hote hain. Common names:

- GoGuardian, Securly, LanSchool, Netop, Impero, ClassDojo, etc.

Ye check karo:

- **Task Manager** → **Processes**: unke processes dikh sakte hain.
- **Services** (`services.msc`): unki services.
- **Startup** (Task Manager → Startup): unki entries.
- **Browser extensions**: Chrome/Edge me extensions check karo.
- **Proxy settings**: `netsh winhttp show proxy`, Internet Options → Connections → LAN settings.
- **Root certificates**: `certmgr.msc` → Trusted Root Certification Authorities. Monitoring tools apna certificate install karte hain HTTPS traffic dekhne ke liye.
- **Hosts file**: `C:\Windows\System32\drivers\etc\hosts`.
- **Firewall rules**: `wf.msc`.

Distributed Shared Memory & Remote Memory Pooling

Repository	Description & Key Features
pojntfx/r3map	High-performance remote memory region mounts and migrations in user space using Go.
uclasystem/MemLiner	Remote-memory-friendly runtime system for memory-disaggregated clusters.
uclasystem/canvas	Isolated and adaptive swapping for multi-applications on remote memory.
uclasystem/Mako	Low-pause, high-throughput garbage collector designed for memory-disaggregated datacenters.
dopecoder/rmpool	C++ remote memory pooling mechanism across network-connected machines.
PUMA	Kernel-level remote memory caching framework for datacenter VMs.
habanero-rice/hclib	Task-based parallel programming model supporting shared and distributed memory.

Memory Disaggregation & Swap Systems

Repository	Description & Key Features
SymbioticLab/Infiniswap	Efficient memory disaggregation via RDMA-based decentralized swap paging.
clusterfarmem/fastswap	Fast swap system for far memory over high-speed RDMA networks.
clusterfarmem/cfm	Cluster far-memory framework complementing fastswap-driven disaggregation.

CXL (Compute Express Link) Memory Pooling

Repository	Description & Key Features
TianheMICALab/SimCXL	Full-system cycle-level simulator based on gem5 supporting all CXL sub-protocols and device types.
cxl-reskit/cxl-reskit	CXL Memory Resource Kit (CMRK) containing diagnostics, benchmarks, and management utilities.

RAM Disk & Block-Level Memory Caching

Repository	Description & Key Features
pkoutoupis/rapiddisk	Advanced Linux RAM drive and caching kernel modules with RESTful API and NVMe Target support.

Kernel-Level Memory Access & DMA Analysis

Repository	Description & Key Features
ri-char/rwMem	Kernel module to read/write memory and set hardware breakpoints in remote processes.
ufrisk/LeechCore	Physical memory acquisition library & remote acquisition agent for hardware and software sources.
ufrisk/pcileech	Direct Memory Access (DMA) attack and memory acquisition software over PCIe and network.